TRACK: UX/UI DESIGN · WEEK 5/7 · BUILD+

UX-07: Designing the AI Surface (Chat, Agents, and Trust)

Proactive Agent Architecture · 4 Shape of AI Patterns · Uncertainty Failure Mode · Microsoft HAX Audit

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Project: FlyCommerce OS

INTERACTION MODEL	SHAPE OF AI PATTERNS	FAILURE MODE	HAX AUDIT SCORE
Autonomous Agent + Human Loop	2 Governors · 2 Trust Builders	Split-Scenario Anomaly Branching	6 Passes · 2 Honest Fails (Fixed)

1. Interaction Model Justification (Agent vs. Chatbot vs. Autopilot)

Selected Model: Proactive Surveillance Agent with Staged Human Review.

• *Why Chatbots Were Rejected:* Natural language typing during a high-stakes stockout crisis adds cognitive latency (>2,000ms), ambiguity in SKU syntax, and buries multi-channel numbers in wordy chat bubbles.

• *Why Full Autopilot Was Rejected:* 68% of operators abandon automated tools that reallocate stock or issue purchase orders without explicit consent (UX-02 Finding 2).

2. Four Named Patterns from "The Shape of AI" Library

GOVERNOR PATTERN

1. Action Plan (Staged Execution)

Decomposes AI proposals into an explicit 3-step execution plan (Step 1: Reserve 150u from Wholesale; Step 2: Push TikTok lock; Step 3: Transmit 3PL pick manifest). Operators can toggle individual steps before execution.

GOVERNOR PATTERN

2. Interactive Verification Slider

Tactile reallocation slider with hard safety locks. If the user drags past 75% of total wholesale reserve, the system stops and requires an explicit checkbox confirmation to prevent accidental stock depletion.

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3. Explainable Formula Disclosure

Replaces black-box "Smart AI" badges with an inspectable math formula: `(Velocity [14.2u/h] × 48h Buffer) + Lead Time = 136 Units`, directly addressing operator skepticism.

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4. Telemetry Citations & Caveats

Cites data recency for every number (Shopify sync 42s ago, TikTok 7d velocity) alongside active supplier caveats: "⚠️ *Factory Blackout in 18 days for Lunar New Year (14d buffer appended)*".

3. AI Failure & Uncertainty Mode: Split-Scenario Branching

⚠️ Anomaly Detected: 16x Viral Sales Surge (Confidence: 48% — Below Automated Action Threshold)

The AI cannot determine if this surge is transient or permanent. Rather than failing silently or throwing a red error toast, it surfaces two divergent scenario models:

Scenario A: Transient Flash Spike
• Rebalance 150 buffer units from wholesale pool.
• Do NOT issue international supplier PO.
Capital at Risk: $0 (Internal Buffer Shift)

Scenario B: Sustained Demand Shift
• Rebalance 300 units from FBA hub.
• Issue emergency Air-Freight Supplier PO.
Capital at Risk: $24,500 (Production Commit)

4. Microsoft HAX Guidelines Audit (Human-AI Interaction)

HAX Guideline	Status	Implementation Evidence in FlyCommerce OS	Audit Finding & Planned Fix
G1: Make clear what system can do	**PASS**	Onboarding modal scopes jurisdiction strictly to inventory monitoring and buffer planning.	Clear system mental model.
G2: Make clear how well it can do it	**PASS**	Every recommendation explicitly displays its confidence score (e.g. "*94% Confidence*").	Prevents over-reliance on AI.
G4: Show contextually relevant info	**PASS**	Telemetry (velocity, lead times, factory blackouts) shown directly adjacent to interaction.	Eliminates menu navigation.
G8: Support efficient dismissal	**PASS**	Proactive warning toasts have an instant single-click 4-hour snooze button.	Preserves user attention.
G9: Support efficient correction	**PASS**	Tactile allocation slider enables instant override of AI-suggested units in <2s.	Zero friction correction.
G11: Make clear why system acted	**PASS**	Pattern 3 (Explainable Formula Drawer) exposes exact arithmetic formula.	Full algorithmic transparency.
G10: Scope services when in doubt	**HONEST FAIL**	Early prototypes computed exaggerated reorders during viral spikes, confusing users.	**Planned Fix Implemented:** Split-Scenario Uncertainty Modal triggered when confidence <70%.
G13: Learn from user behavior	**HONEST FAIL**	If Elena repeatedly overrides the 15% buffer to 25%, the system forgot her preference.	**Planned Fix Implemented:** Adaptive Memory Token prompts default update after 3 overrides.

Master Figma Workspace (View Access): https://www.figma.com/file/r7K8x9pL4mN2qS/FlyRank-UX-UI-Design-Capstone?type=design&node-id=0%3A1&mode=design&t=review-access
GitHub Repository: https://github.com/KMSiam/Flyrank-Internship-UX-UI-Design